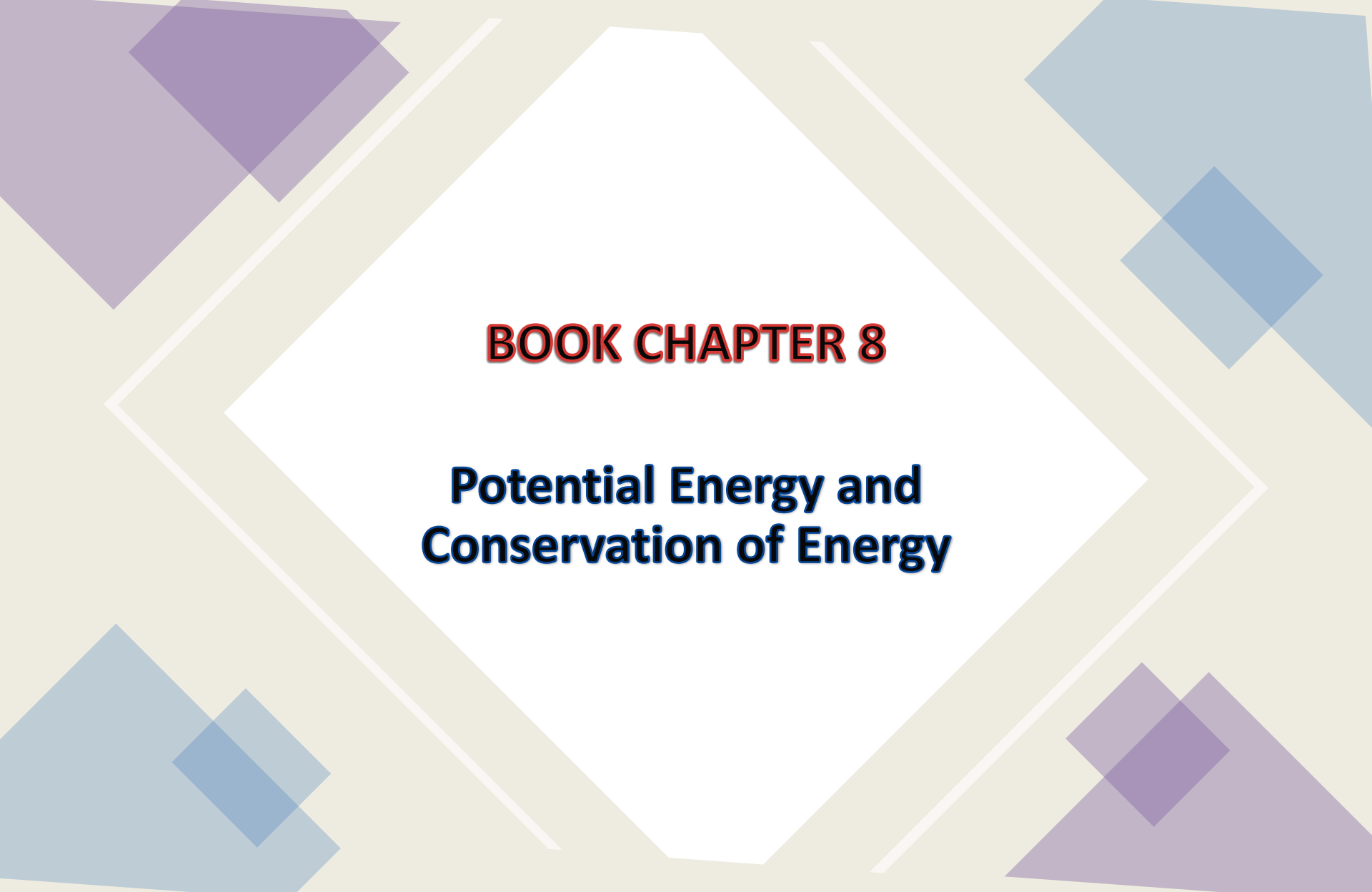


LESSON 7



- **BOOK CHAPTER 8**
- **Potential Energy and Conservation of Energy**



BOOK CHAPTER 8

Potential Energy and Conservation of Energy

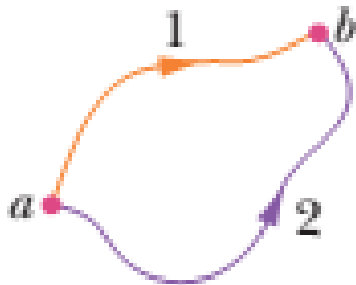
Conservative and non-conservative forces:

Conservative force:

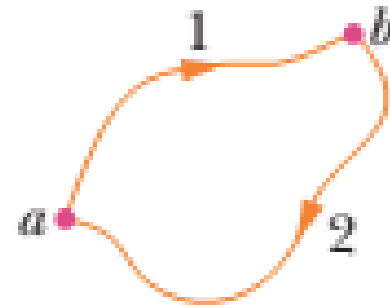
A force is a **conservative force** if the net work it does on a particle moving around any closed path, from an initial point and then back to that point, is zero.

Equivalently, **a force is conservative** if the net work it does on a particle moving between two points does not depend on the path taken by the particle. **Examples:** The **gravitational force** and the **spring force** are conservative forces.

The term conservative force comes from the fact that when a conservative force exists, it conserves mechanical energy.

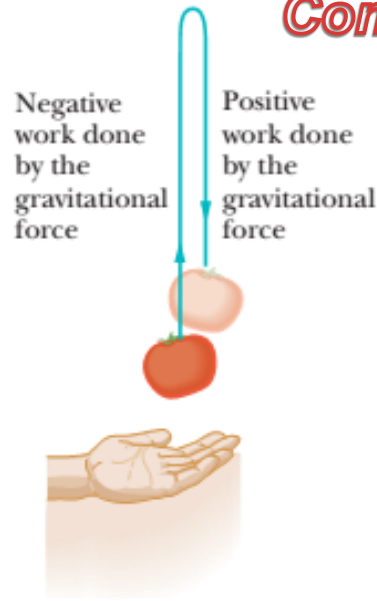


The force is conservative. Any choice of path between the points gives the same amount of work.

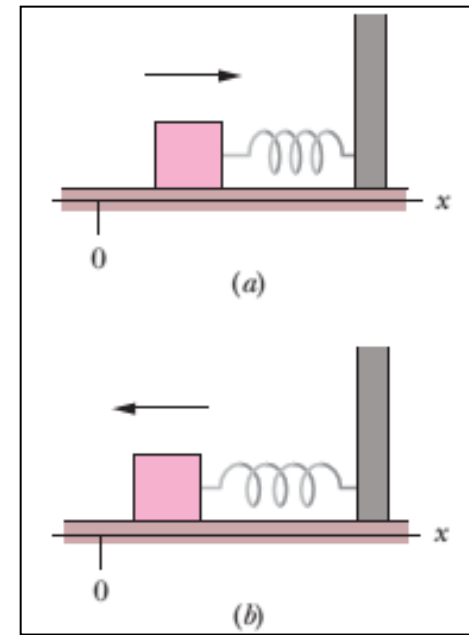


And a round trip gives a total work of zero.

Conservative force:



A tomato is thrown upward. As it rises, the gravitational force does negative work on it, decreasing its kinetic energy. As the tomato descends, the gravitational force does positive work on it, increasing its kinetic energy.



A block, attached to a spring and initially at rest at $x = 0$, is set in motion toward the right. (a) As the block moves rightward (as indicated by the arrow), the spring force does negative work on it. (b) Then, as the block moves back toward $x = 0$, the spring force does positive work on it.

For either rise or fall, the change ΔU in gravitational potential energy is defined as being equal to the negative of the work done on the tomato by the gravitational force. Using the general symbol W for work, we write this as

$$\Delta U = -W$$

Non-conservative force:

A force that is not conservative is called a **non-conservative force**. The **kinetic frictional** force and **drag force** are non-conservative.

For an example, let us send a block sliding across a floor that is not frictionless. During the sliding, a kinetic frictional force from the floor slows the block by transferring energy from its kinetic energy to a type of energy called *thermal energy* (which has to do with the random motions of atoms and molecules). We know from experiment that this energy transfer cannot be reversed (thermal energy cannot be transferred back to kinetic energy of the block by the kinetic frictional force).

The principle of Conservation of Mechanical Energy:

The mechanical energy E_{mec} of a system is the sum of its kinetic energy K and potential energy U of the objects within it. That is,

$$E_{mec} = K + U$$

This conservation principle can also be written as

$$\Delta E_{mec} = \Delta K + \Delta U = 0$$

Here $\Delta K = +W$

and $\Delta U = -W$

Let's try.....

1. [Chap 8 - problem 1]: What is the spring constant of a spring that stores 25 J of elastic potential energy when compressed by 7.5 cm?
2. [Chap 8 - problem 3]: You drop a 2.00 kg book to a friend who stands on the ground at distance $D = 10.0$ m below. If your friend's outstretched hands are at distance $d = 1.50$ m above the ground (Fig. 8-30), (a) how much work W_g does the gravitational force do on the book as it drops to her hands? (b) What is the change ΔU in the gravitational potential energy of the book–Earth system during the drop? If the gravitational potential energy U of that system is taken to be zero at ground level, what is U (c) when the book is released and (d) when it reaches her hands? Now take U to be 100 J at ground level and again find (e) W_g , (f) ΔU , (g) U at the release point, and (h) U at her hands.
3. [Chap 8 - problem 10]: (a) In Problem 3, what is the speed of the book when it reaches the hands? (b) If we substituted a second book with twice the mass, what would its speed be? (c) If, instead, the book were thrown down, would the answer to (a) increase, decrease, or remain the same?

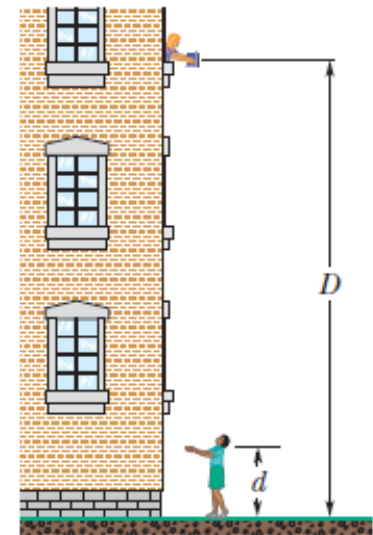


Figure 8-30
Problems 3 and 10.



**Thank
You**